STAT 4830 - SPRING 2026

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Optimizing Hyperparameter Tuning Methods for *Evolutionary Strategies*

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Problem Statement

Problem Statement

- **Goals:**
  1. Continue with **population-scale/Top-K** experiments
  2. Establish **baseline model accuracy** for post-trained, pre fine-tuned models on different model sizes and tasks
  3. Explore the Neural Thickets method and establish a baseline

02
Technical Approach

Technical Approaches - ES

Algorithm 1 Basic ES Algorithm

Require: Pretrained LLM with initial parameters $\boldsymbol{\theta}_0$, reward function $R(\cdot)$, total iterations T, population size N, noise scale σ, learning rate α.

for $t = 1$ to T **do** ▷ outer ES iterations
  for $n = 1$ to N **do**
    Sample noise $\boldsymbol{\varepsilon}_n \sim \mathcal{N}(0, \boldsymbol{I})$
    Compute reward for perturbed parameters: $R_n = R(\boldsymbol{\theta}_{t-1} + \sigma \cdot \boldsymbol{\varepsilon}_n)$
  end for
  Normalize R_n
  Update model parameters as $\boldsymbol{\theta}_t \leftarrow \boldsymbol{\theta}_{t-1} + \alpha \cdot \frac{1}{N} \sum_{n=1}^{N} R_n \boldsymbol{\varepsilon}_n$
end for

Sample Perturbations

Evaluate Perturbed Models

Update Parameters

Qiu, Xin, et al. "Evolution Strategies at Scale: LLM Fine-Tuning Beyond Reinforcement Learning." ArXiv.org, Sept. 2025, https://doi.org/10.48550/arxiv.2509.24372.

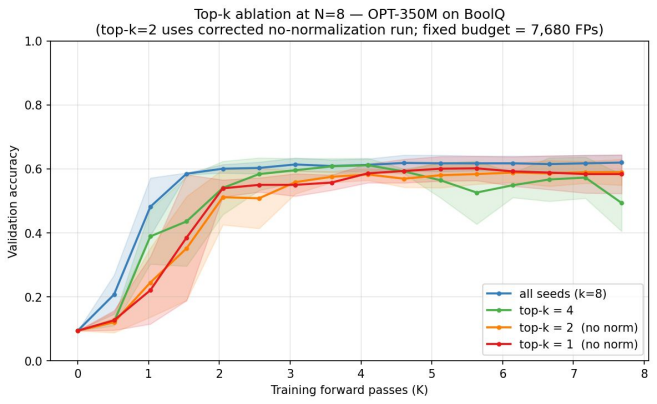
03

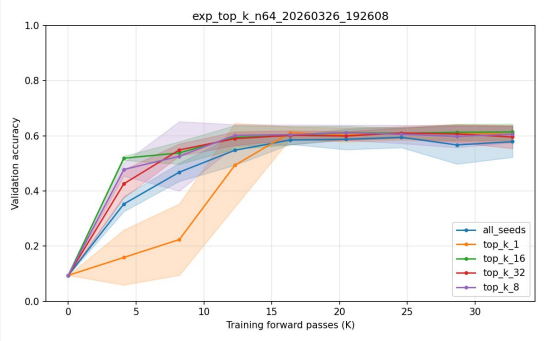
Results

Results 1 - Population/Top-K

Continuation of population-scaling/Top-K experiments
Fixed forward pass budget at K = 7680

- OPT 350M on BoolQ (TF questions)
  - For k ≤ 2, ran with normalization disabled
  - **Key Finding:** Top-K might not have a meaningful effect on accuracy convergence





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- OPT 350M on BoolQ (TF questions)
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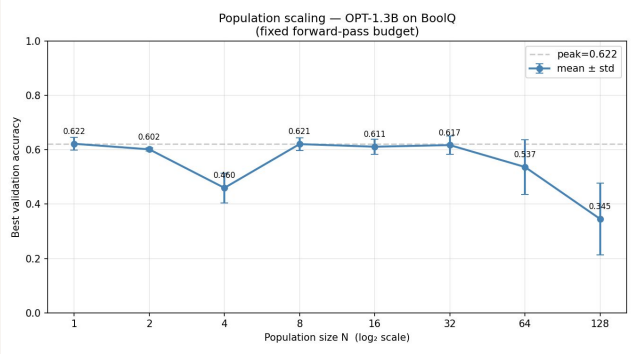
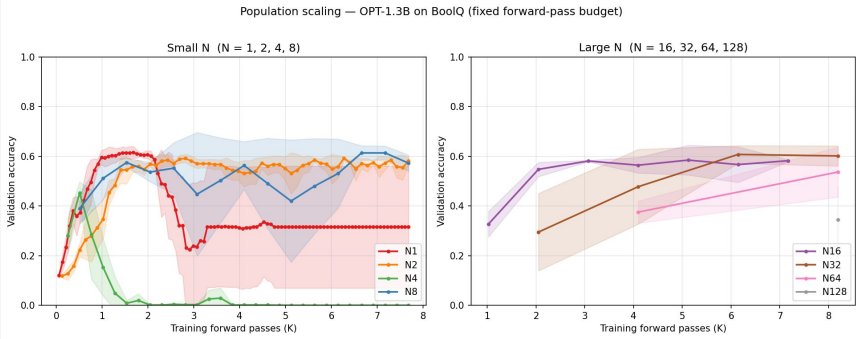
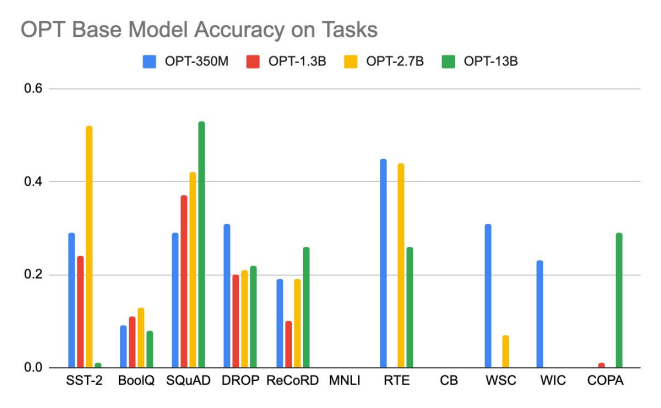
  - **Key Finding:** Top-K might not have a meaningful effect on accuracy convergence
- OPT 1.3B on BoolQ
  - Rise-then-decay shape
  - We exhaust our budget for N=64/128

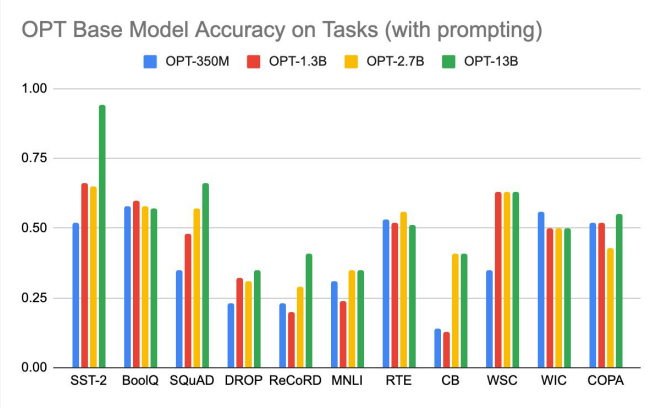




Results 2 - Baseline for Hyperparameter tuning

- **Key finding:** a lot of "parse failures" for language inference tasks like MNLI, CB, WSC, COPA
- Noticed a lot more parse failures in OPT-13B. Bottom plot shows after changing max-token from 4 to 16





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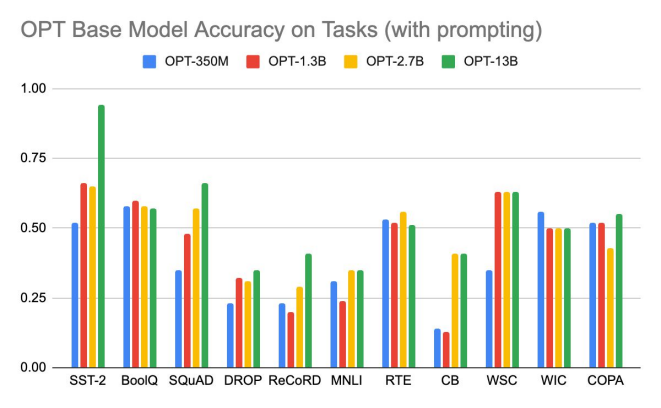

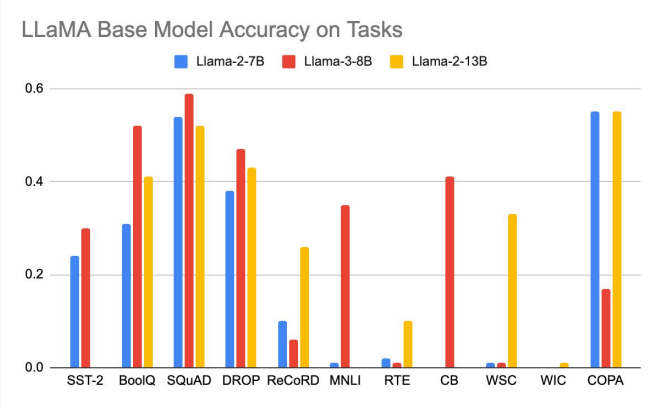

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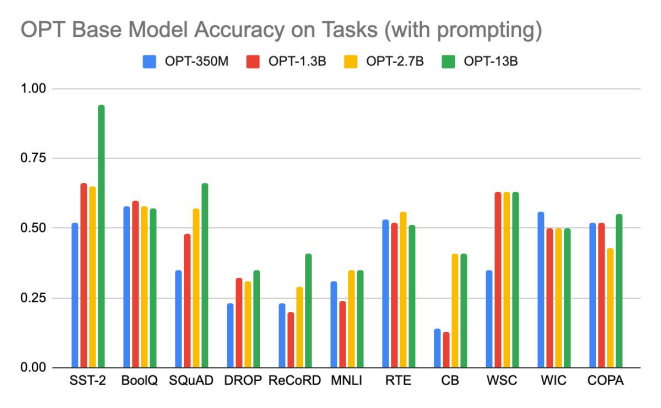
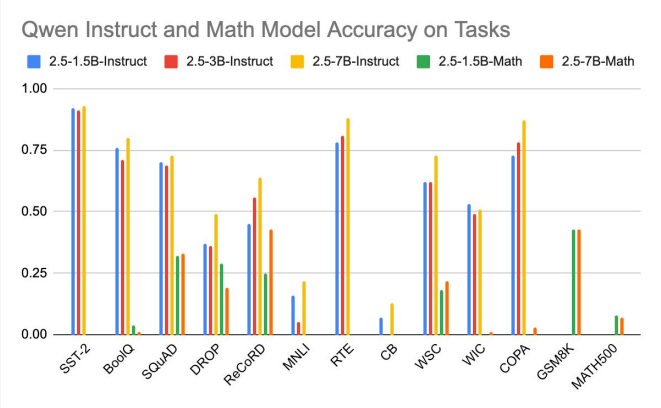
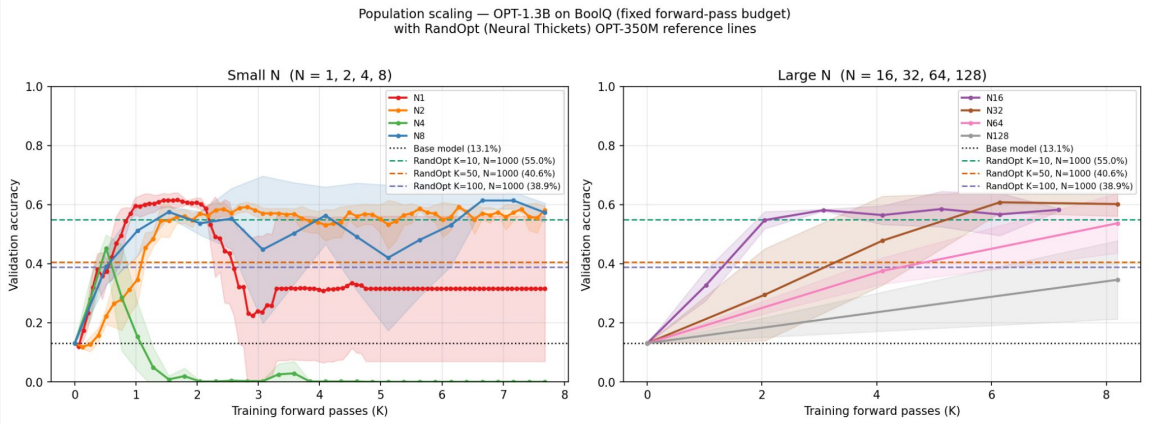




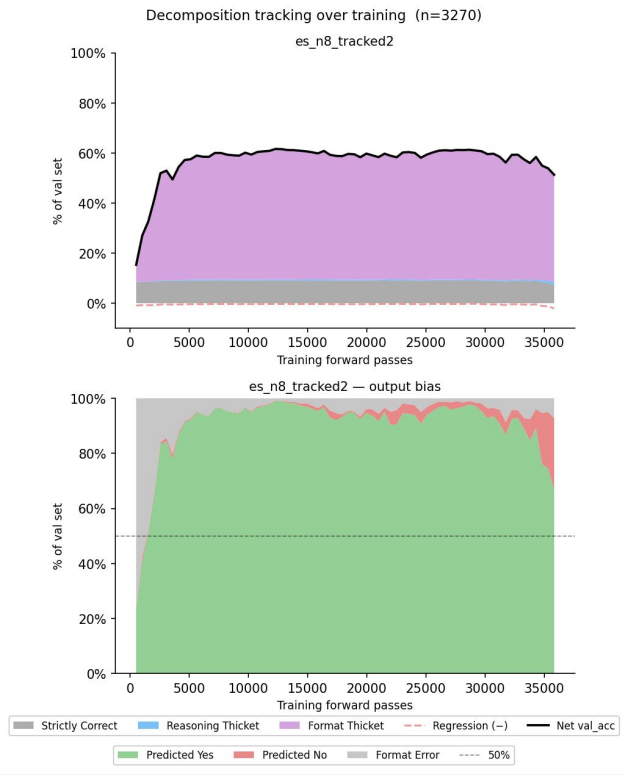
Results 3 - Neural Thickets

- **RandOpt (K=10, N=1000) at ~55.0% matches the best ES curve peak**
  - comparable gain over base model (13.1%) without iterative training
- **The ES uptick and RandOpt's one-shot improvement land at the same forward-pass count (~1K)**
  - per-forward-pass efficiency is equivalent
- **However, both of them seemingly exhibit reward-hacking behavior (cont'd on the next slide)**



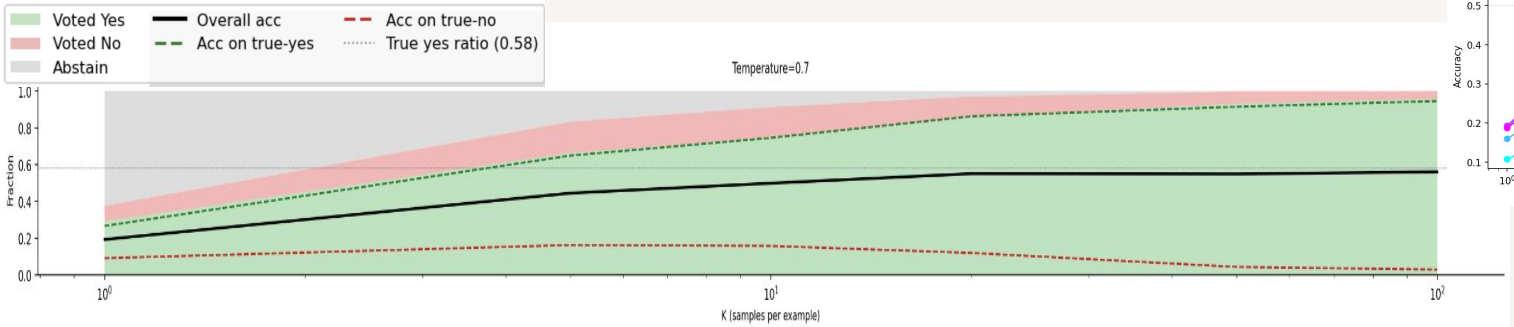
Results 4 - ES "Improvement" on BoolQ is Reward Hacking

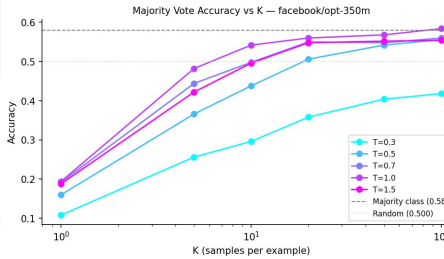
- **ES achieves ~62% accuracy**
  - matching the majority-class baseline (62.2% of BoolQ labels are "yes")
- **Decomposition shows zero Reasoning Thicket**
  - all gain comes from Format Thicket (fixing malformed outputs), not corrected reasoning
- **Output bias panel confirms: model outputs ~90% "yes" across all training steps**
  - it learned to exploit the label skew, not the task



Results 5 - Base Model Sweep (w/o perturbation)

- **K = number of stochastic samples per question**
  - for each val example (~500), the model generates K outputs at temperature T; majority vote over valid yes/no responses gives the final prediction
- **Voted Yes/No/Abstain (stacked area)**
  - fraction of the 500 val examples whose majority vote landed on yes, no, or all-format-error respectively
- **Acc on true-yes / true-no (dashed lines): recall broken down by true label**
  - if acc-true-no ≈ 0, the model never predicts "no" regardless of the passage content
- **Overall acc (black line): net accuracy across all 500 examples**
  - tracks closely with Voted Yes fraction, confirming accuracy gains come from yes-bias not task understanding
- **Even vanilla model without any perturbation shows reward hacking behvaior on many samplings!**





04

Limitations

Limitations

- **Double down on Hyperparameter Scaling Law beyond baselines**
  - We don’t yet have reasonable values for some hyperparameters (perturbation scale, learning rate) to run our full experiment
  - Baseline experiment is not complete
- **Need more complex task & larger models to see if ES actually improves reasoning capabilities**
  - OPT350M & BoolQ was too reward-hackable

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Next Steps

Next Steps

- **For BoolQ convergence independent of N,** continue experiments for different model sizes, tasks. Explore answer parsing and possible model hacking
- **For baseline accuracy study,** continue testing with more models and tasks
- **For Neural Thickets & Base Model Sweep**, run bigger model on more complex tasks to prevent reward hacking and compare a real reasoning improvement vs. ES

THANK YOU

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